

Annotated Plots

```
In [40]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from matplotlib.colors import ListedColormap
```

```
In [41]: df = pd.read_csv("data.csv")
df.head()
```

```
Out[41]:
```

	tech_company	benefits	workplace_resources	mh_employer_discussion	mh_cowor
0	Yes	No	I don't know		No
1	Yes	Yes	No		No
2	Yes	I don't know	No		Yes
3	Yes	Yes	I don't know		Yes
4	Yes	Yes	No		No

```
In [42]: gender = df["gender"].value_counts()

colors = []
for g in gender.index:
    if g == "Male":
        colors.append("#4682B4")
    elif g == "Female":
        colors.append("#CD5C5C")
    else:
        colors.append("#A9A9A9")

male_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Male')]
female_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Female')]
other_with_mental_health = df[(df["mental_health"] == 'Yes') & (df["gender"] == 'Other')]

sizes = [
    male_with_mental_health['gender'].shape[0],
```

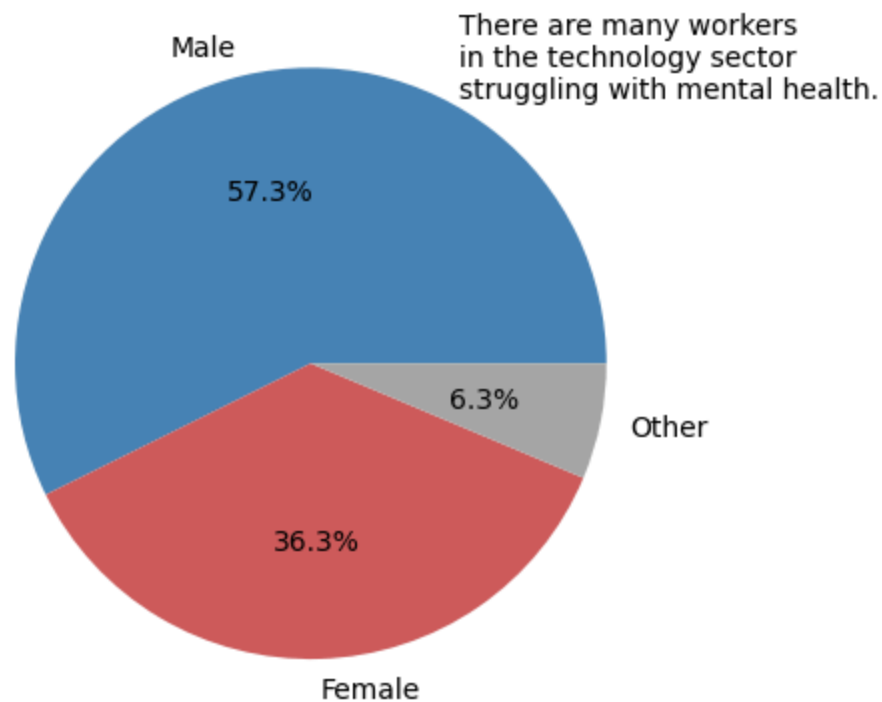
```

female_with_mental_health['gender'].shape[0],
other_with_mental_health['gender'].shape[0]
]
labels = ['Male', 'Female', 'Other']

plt.pie(sizes, labels = labels, autopct = '%1.1f%%', colors=colors)
plt.annotate("There are many workers\nin the technology sector\nstruggling w
plt.title("Mental Health Across Genders in the Technology Industry")
plt.show()

```

Mental Health Across Genders in the Technology Industry

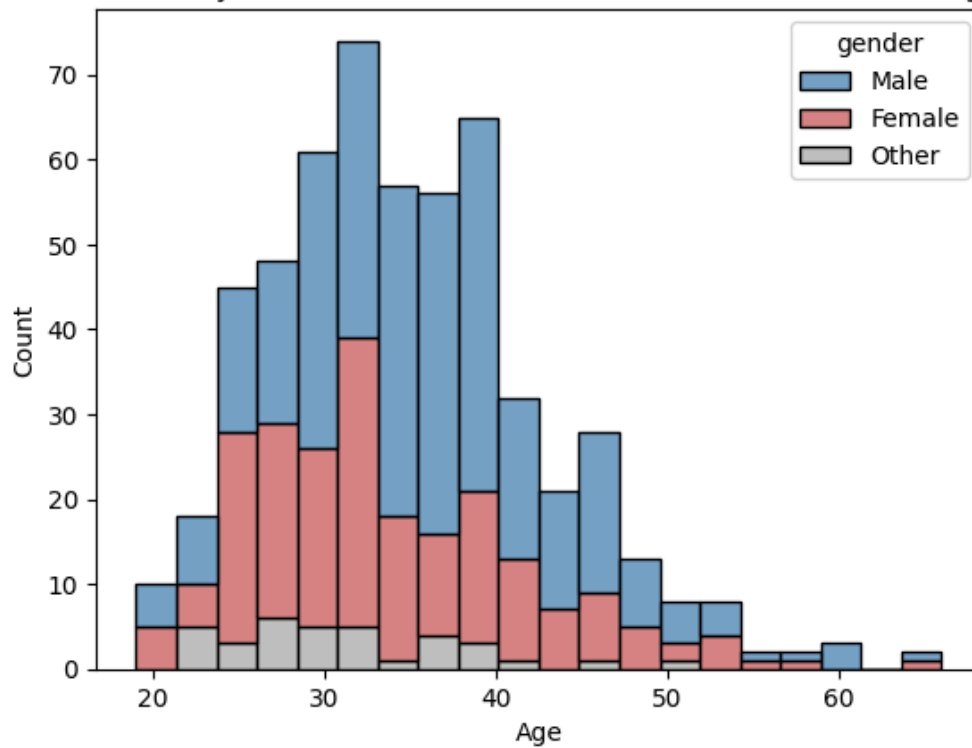


```

In [43]: gender = df[df["mental_health"] == "Yes"]
sns.histplot(gender, x="age", hue="gender", multiple="stack", palette={"Male
plt.xlabel("Age")
plt.title("Age Distribution By Gender Who Have Mental Health in the Technolo
plt.show()

```

Age Distribution By Gender Who Have Mental Health in the Technology Industry

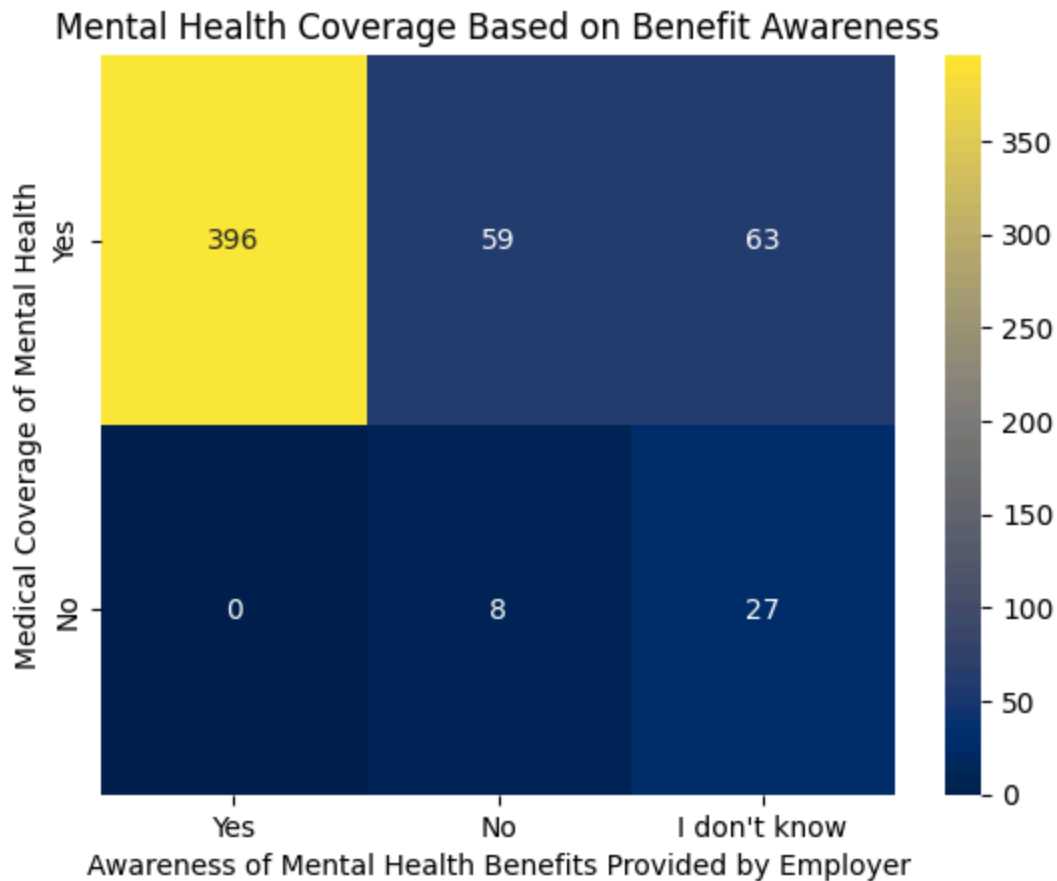


```
In [44]: df_mh = df[df["mental_health"] == "Yes"].copy()

df_mh["benefits"] = pd.Categorical(df_mh["benefits"], categories=["Yes", "No"])
df_mh["medical_coverage"] = pd.Categorical(df_mh["medical_coverage"], categories=["Yes", "No"])

heat = pd.crosstab(df_mh["medical_coverage"], df_mh["benefits"])

sns.heatmap(heat, annot=True, cmap="cividis", fmt="d")
plt.xlabel("Awareness of Mental Health Benefits Provided by Employer")
plt.ylabel("Medical Coverage of Mental Health")
plt.title("Mental Health Coverage Based on Benefit Awareness")
plt.show()
```



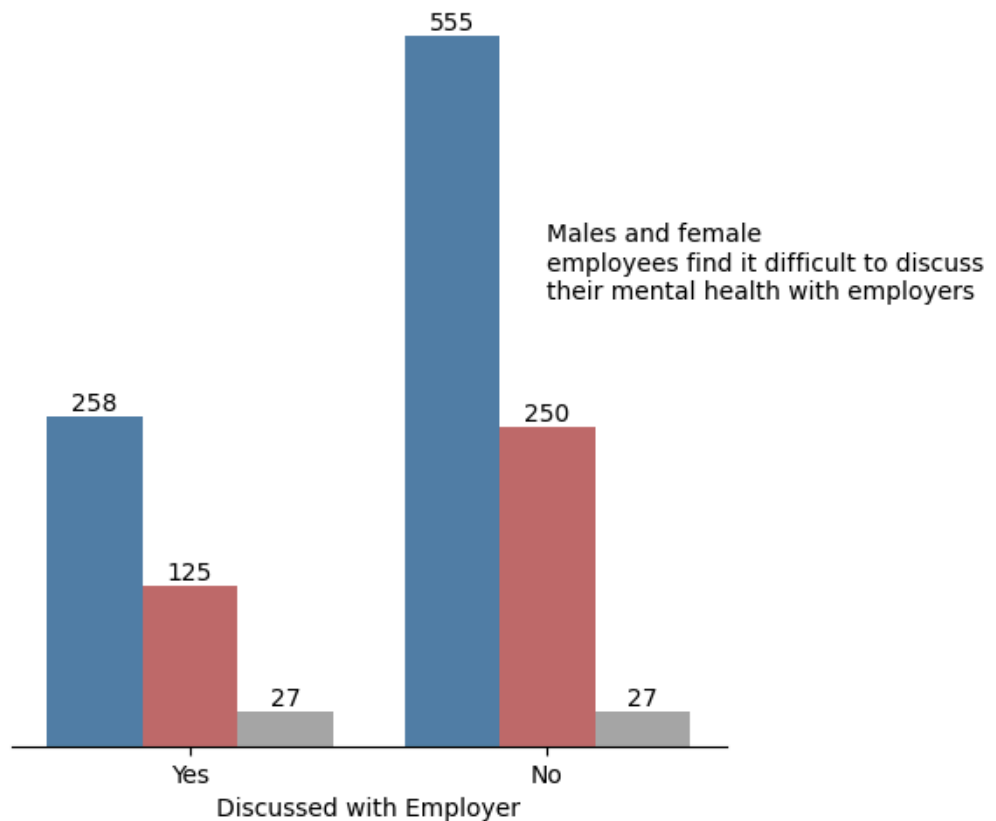
```
In [45]: discussion = df.groupby("mh_employer_discussion")["gender"].value_counts().r
discussion_graph = sns.catplot(discussion, x="mh_employer_discussion", y="Co

discussion_graph.set_axis_labels("Discussed with Employer", "")
for ax in discussion_graph.axes.flat:
    ax.spines['left'].set_visible(False)
    ax.set_yticks([])

plt.title("Mental Health Discussion with Employer By Gender in the Technolog
discussion_graph._legend.remove()
for ax in discussion_graph.axes.flat:
    for p in ax.patches:
        height = p.get_height()
        if height > 0:
            ax.text(
                p.get_x() + p.get_width()/2.,
                height + 0.5,
                int(height),
                ha='center',
                va='bottom'
            )

plt.annotate("Males and female\nemployees find it difficult to discuss\nthei
plt.show()
```

Mental Health Discussion with Employer By Gender in the Technology Industry

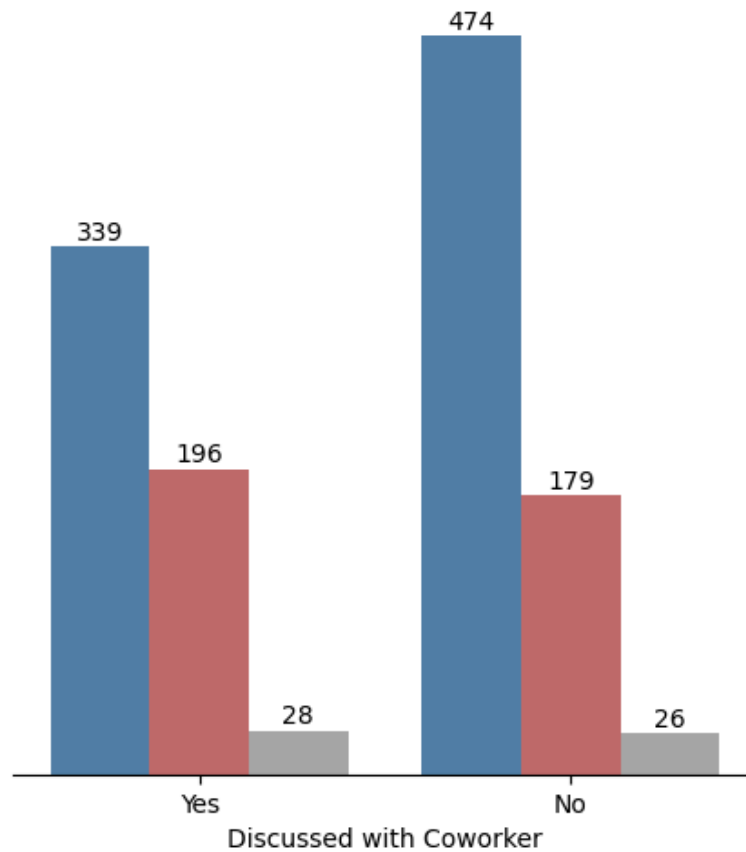


```
In [46]: discussion2 = df.groupby("mh_coworker_discussion")["gender"].value_counts().
discussion2_graph = sns.catplot(discussion2, x="mh_coworker_discussion", y="
discussion2_graph.set_axis_labels("Discussed with Coworker", "")
for ax in discussion2_graph.axes.flat:
    ax.spines['left'].set_visible(False)
    ax.set_yticks([])

plt.title("Mental Health Discussion with Coworker By Gender in the Technolog
discussion2_graph._legend.remove()
for ax in discussion2_graph.axes.flat:
    for p in ax.patches:
        height = p.get_height()
        if height > 0:
            ax.text(
                p.get_x() + p.get_width()/2.,
                height + 0.5,
                int(height),
                ha='center',
                va='bottom'
            )

plt.show()
```

Mental Health Discussion with Coworker By Gender in the Technology Industry



```
In [47]: df["gender_color"] = df["gender"].map({"Male": 0, "Female": 1, "Other": 2})

plot = px.parallel_categories(df, dimensions=["gender", "mental_health", "wc"],
                             labels={"gender": "Gender Identity", "mental_h": "Mental Health Response", "wc": "Knowledge of Workplace Resources"},
                             color="gender_color", color_continuous_scale=px.colors.qualitative.M3Sequential)
plot.update_coloraxes(showscale=False)
```

